CELESTIAL HORIZONS

EXPLORE THE UNCHARTED INFINITY

Our mission: to design and build a compact 101 cm experimental rocket.

Our system must withstand extreme acceleration, vibrations, and engine heat while complying with very limited internal volume requirements.

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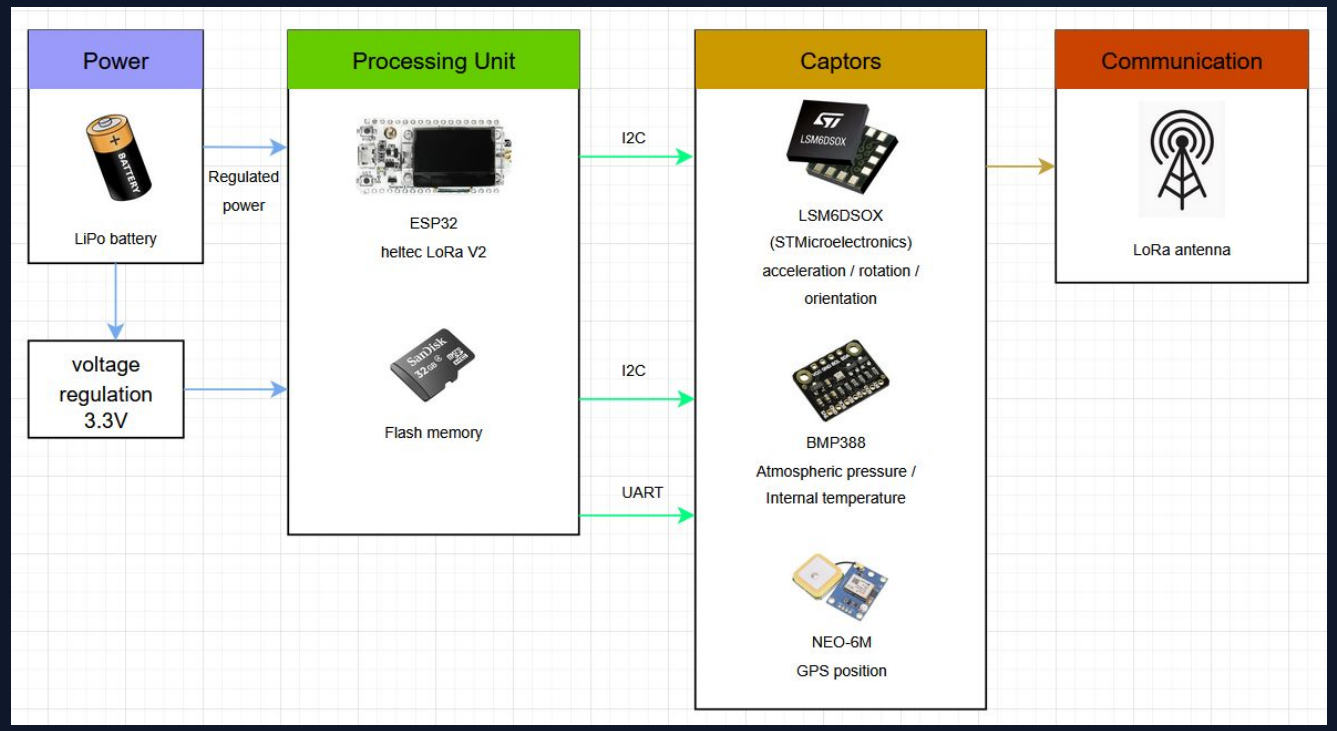
APOLLO NETWORK

DEEP SPACE DISCOVERY

Architecture & Sensors

- System Core: ESP32 Heltec LoRa V2 powered by a LiPo battery to ensure centralized processing.
- Inertial Unit: LSM6DSOX sensor measuring acceleration, rotation, and orientation at high frequency (up to 200 Hz).
- Environment: BMP388 barometer for accurate altitude estimation and internal temperature monitoring.
- Location: NEO-6M GPS module to transmit exact coordinates and facilitate recovery.



The 5 Phases of Flight

IDLE
Waiting at the launch pad in low-power mode.

BOOST
High acceleration, IMU sampling at 200 Hz.

FLIGHT
Free flight, altitude and stability calculation.

DESCENT
Barometric tracking of the parachute trajectory.

LANDING
GPS activation and LoRa data transmission.

LoRa communication

During the Flight

Minimalist Approach

To avoid saturating bandwidth, we use a cautious strategy:

- Sending a simple beacon every 10 seconds.
- Tag content: Altitude, timestamp, battery status.
- Heavy data (IMU) is stored locally and not transmitted.

After Landing

Core Strategy

The transmission engages fully once on the ground, in two stages:

- **1. Immediate Summary: Maximum altitude, peak acceleration, and GPS coordinates for recovery.**
- **2. Data Dump: Complete dump of history over 2 to 3 minutes (pressure, temperature, rotation).**

Energy Management

LiPo battery autonomy is optimized via ESP32 modes and sensor adaptation. Voluminous data is compressed (50-byte packets) and secured (CRC, acknowledgment).